WG 7 - Francesca

Intelligent Systems

Exercise 1: Design Matrixes

Consider the following training set of positive examples {(1,1),(1,-1),(-2,2),(-2,-2)} and negative examples {(3,3),(3,-3),(-4,4),(-4,-4)}. Draw the feature space and try to fit a linear separator between the classes.

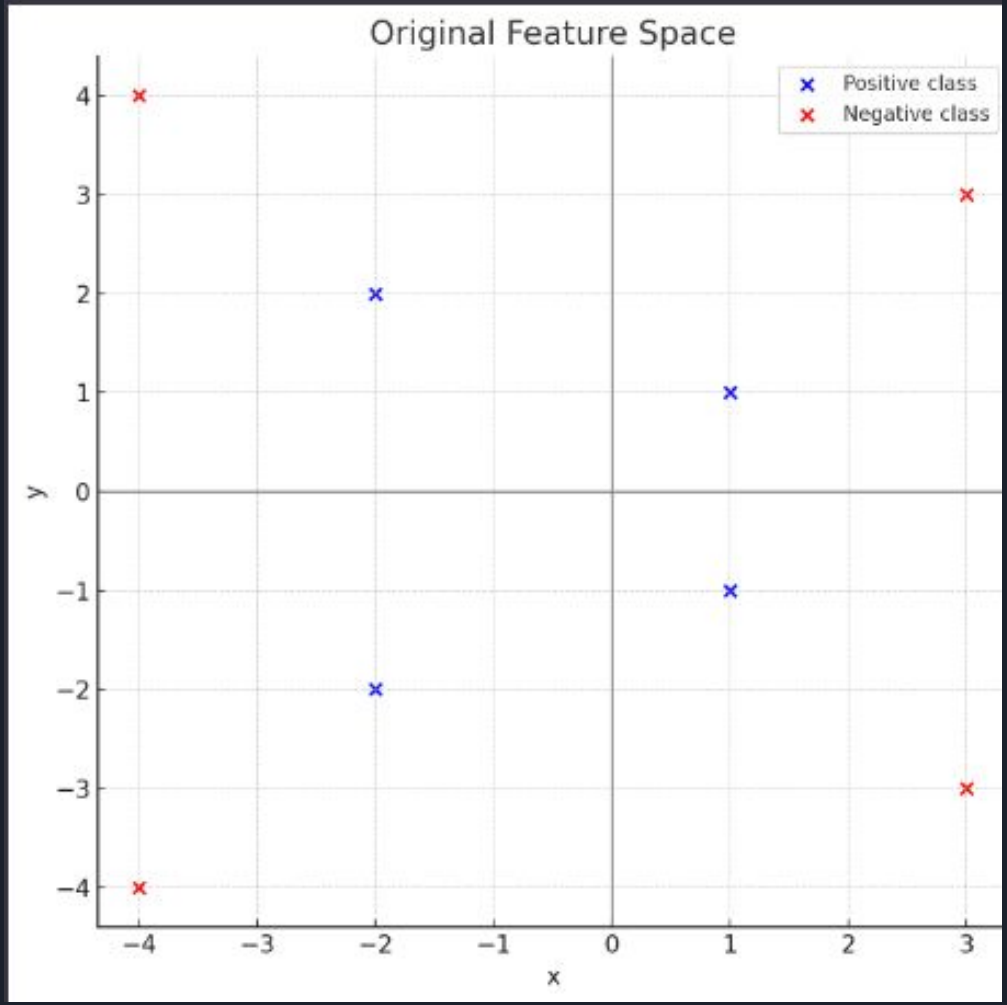
What can you do to rectify the problem? Provide a simple design matrix with two combined features instead of the original ones for which there is a simple separator, and give an example for such a linear separator.

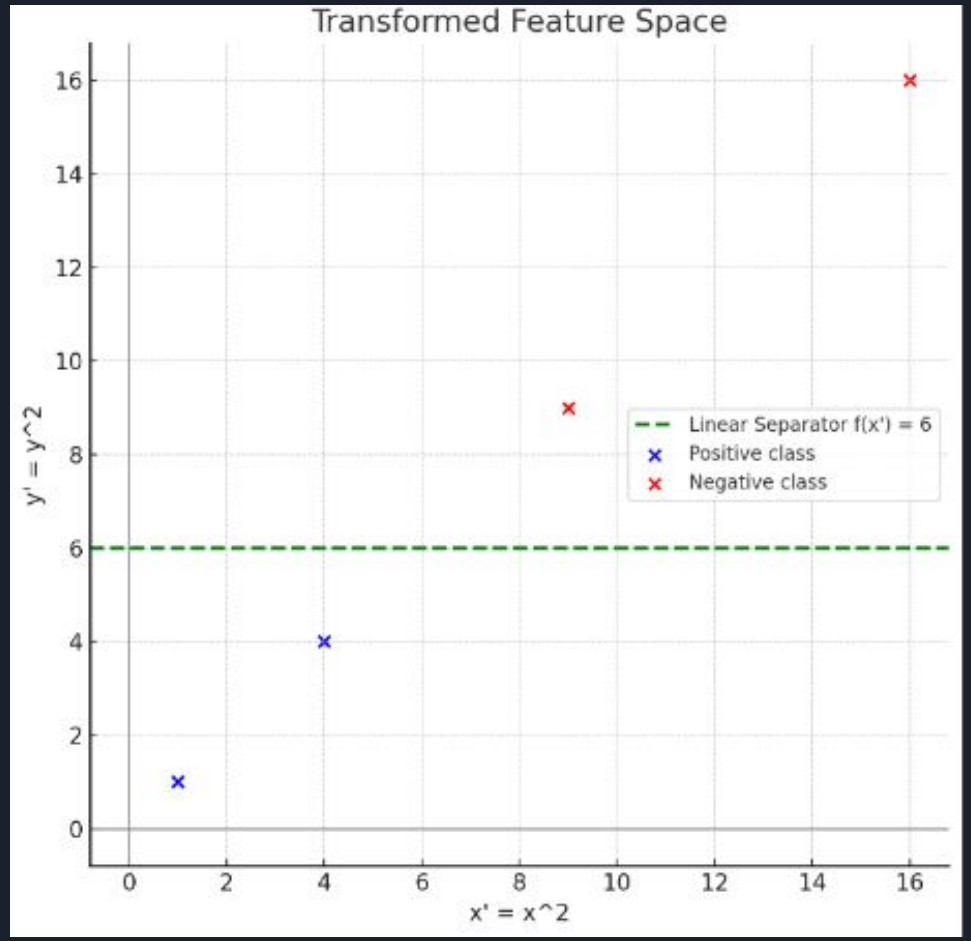
PROBLEM

In the original feature space, the classes are not linearly separable, but by transforming the features into a new space, we can find a linear boundary that separates the two classes.

positive examples {(1,1),(1,-1),(-2,2),(-2,-2)}

negative examples {(3,3),(3,-3),(-4,4),(-4,-4)}.





If you choose two features $x' = x^2$ and $y' = y^2$ you get a new positive training set {(1,1),(4,4)} and a negative one {(9,9),(16,16)}.

Any line in between those two sets is a linear separator, e.g. $f(x') = 6$.

Exercies 2: Linear Regression

Here we are going to practise fitting a linear function to a number of training sets. Suppose we have three pairs of values (1,1), (2,3) and (3,2) for two features. As discussed in the lecture, according to the ML recipe, the results depend on the chosen model class. Let us calculate the optimal function according to a number of different classes.

2a) Let us start with the most simple model class to fit our training examples, a constant line. This means that we are looking at a function f(x) = b, which is a horizontal line in the coordinate system. What is the result of learning the optimal function f(x) from the given data? Hint: we are looking for a value for b.

Solve this problem by using the following steps:

1. Calculate the value of the (squared) error function for each of the three points
2. Calculate the error function based on the points for the three points
3. Search for the minimal value for this function, by
    a. Calculating the derivation of the function w.r.t. b
    b. Setting this derivation to 0, and solve the equation for b.

SOLUTION

STEP 1 (error function):

$E(b) = (b-1)^2 + (b-3)^2 + (b-2)^2$

STEP 2 (Minimizing error function):

Take derivative of $E(b) = 2(b-1) + 2(b-3) + 2(b-2) = 6b - 12$

STEP 3 (Set derivative equal to 0, solve and find b!)

$6b - 12 = 0$

$b = 12 / 6 = 2$

2b) A slightly more complex example is a line with a given, constant slope. Here we choose e.g. s=1. This means that we are looking at a function f(x) = 1*x + b, which is a line in the coordinate system with a slope of 45 degree. What is the result of learning the optimal function f(x) from the given data? Hint: we are still only looking for a value for b.

Follow the same method as described for 2a.

SOLUTION

STEP 1 (error function):

$E(b) = (1 + b - 1)^2 + (2 + b - 3)^2 + (3 + b - 2)^2$

$= b^2 + (b-1)^2 + (b+1)^2$

$= 3b^2$

STEP 2 (Minimizing error function):

Take derivative of $E(b) = 6b$

STEP 3 (Set derivative equal to 0, solve and find b!)

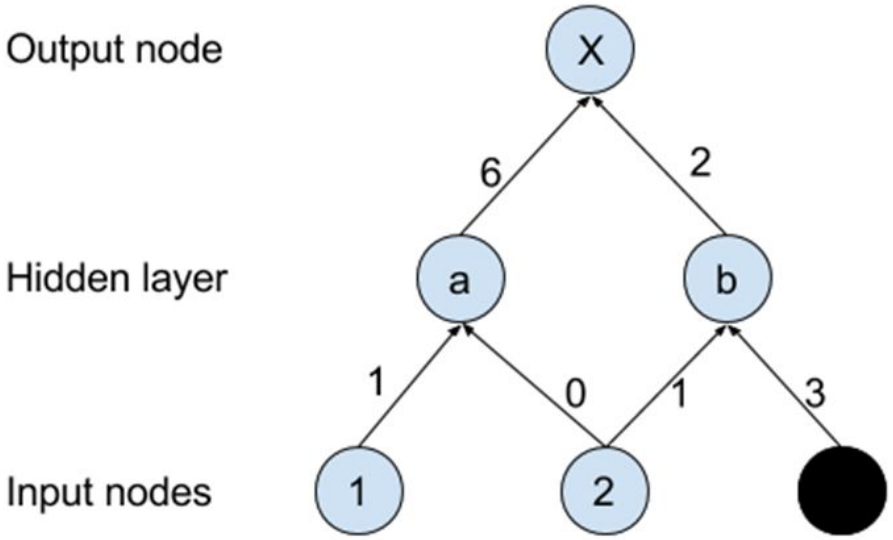
$6b = 0$

$b = 0$

THE FUNCTION $f(x) = x + 0 = x$ IS THE BEST OPTIMAL FITTING FUNCTION FOR OUR SET

Exercise 3: Neural Networks

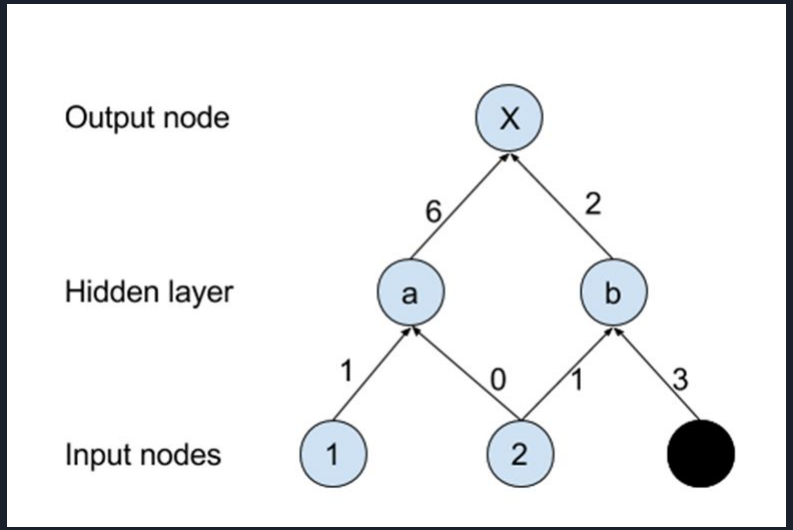
Here, we see a basic multilayer feedforward neural network, with weights and values on the input nodes. It has no activation function (all nodes are *linear*). The black node is a *bias node*. What is the value on the output node (marked X)?



GENERAL FORMULA TO COMPUTE THE VALUE OF A NODE

y= w1*x1 + w2*x2

x values → edges
weights → value inside node



a = 1*1 + 2*0 = 1

b = 2*1 + 1*3 (bias node always 1) = 5

X = 1*6 + 5*2 = 16

ACTIVATION FUNCTION

A mathematical gate in between the input feeding the current neuron and its output going to the next layer; it decides whether a neuron should be activated or not by calculating a weighted sum and further adding bias with it. Its role is to introduce non-linear properties to the network, allowing it to learn more complex functions than what would be possible with linear activation alone.

SIGMOID FUNCTION

In a neural network, the sigmoid function is often used as an activation function to introduce nonlinearity. It squashes the input values to be within the range (0, 1), which can be interpreted as probabilities or be used for further processing in subsequent layers of a neural network.

EXAMPLE

Imagine you have a single feature: the number of times the word "free" appears in an email. You decide to use a sigmoid function to classify the emails.

1. You assign a weight to the feature (let's say `0.8`) and a bias (let's say `-3`).
2. When you receive an email, you count the number of times "free" appears. Suppose it appears 5 times.

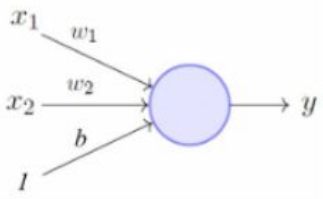
Now, you apply the sigmoid function:

- First, you calculate the weighted sum: $(0.8 \times 5) - 3 = 1$.
- Then, you apply the sigmoid function to this sum: $\frac{1}{1+e^{-1}} \approx 0.73$.

Since the sigmoid output is closer to 1 than to 0, you could classify this email as "spam". The sigmoid function has taken a simple count and transformed it into a classification decision.

Exercise 4: Neural Networks 2

The following node in the neural network uses a sigmoid nonlinearity, and its weights and bias are w_1=0.24, w_2=0.56, b=0.24.



What does it output for the input x_1=6.4, x_2=2.2 ?

The weights and bias for the node are given as:

- $w_1 = 0.24$
- $w_2 = 0.56$
- $b = 0.24$

The inputs are provided as:

- $x_1 = 6.4$
- $x_2 = 2.2$

0.64 * 2.4 + 0.56*2.2 + 0.24 = 3.008
1.0/(1.0+exp(- 3.008)) = 0.95

The calculation shown in the image follows these steps:

1. Multiply each input by its corresponding weight 6.4 $* w_1$ and $2.2 * w_2$.
2. Add the results of the multiplication together with the bias 6.4 $* 0.24 + 2.2 * 0.56 + 0.24$.
3. The sum of these calculations is 3.008.
4. The result is then put through the sigmoid function, which is represented by the formula $\sigma(x) = \frac{1}{1+e^{-x}}$, to get the final output y.
5. The sigmoid of 3.008 is calculated as $\frac{1}{1+e^{-3.008}}$, which results in an output of approximately 0.95.

EXAM RECAP

Exercise 4: Fuzzy Logic

Given the following (standard) semantics for Fuzzy Logic for the following operators:

- $f(\neg A) = 1 - f(A)$
- $f(A \vee B) = \max(f(A),f(B))$
- $f(A \& B) = \min(f(A),f(B))$
- $f(A \rightarrow B) = \min(1;1 - f(A) + f(B))$

What would be the fuzzy interpretation of the formula $((P \rightarrow \neg Q) \rightarrow (Q \vee \neg P)$ given fuzzy values $f(P)=0.9$ and $f(Q)=0.2$ for P and Q.

SOLUTION

$f(\neg A) = 1 - f(A)$

$f(A \vee B) = \max(f(A), f(B))$

$f(A \wedge B) = \min(f(A), f(B))$

$f(A \to B) = \min(1, 1 - f(A) + f(B))$

$f(P) = 0.9, f(Q) = 0.2$

Calculate $f(\neg Q)$:

$$f(\neg Q) = 1 - f(Q) = 1 - 0.2 = 0.8$$

Calculate $f(P \to \neg Q)$: Using $f(A \to B) = \min(1, 1 - f(A) + f(B))$,

$$f(P \to \neg Q) = \min(1, 1 - f(P) + f(\neg Q))$$

Substituting values:

$$f(P \to \neg Q) = \min(1, 1 - 0.9 + 0.8) = \min(1, 0.9) = 0.9$$

Calculate $f(\neg P)$:

$$f(\neg P) = 1 - f(P) = 1 - 0.9 = 0.1$$

Calculate $f(Q \vee \neg P)$: Using $f(A \vee B) = \max(f(A), f(B))$,

$$f(Q \vee \neg P) = \max(f(Q), f(\neg P))$$

Substituting values:

$$f(Q \vee \neg P) = \max(0.2, 0.1) = 0.2$$

Calculate $f((P \to \neg Q) \to (Q \vee \neg P))$: Using $f(A \to B) = \min(1, 1 - f(A) + f(B))$,

$$f((P \to \neg Q) \to (Q \vee \neg P)) = \min(1, 1 - f(P \to \neg Q) + f(Q \vee \neg P))$$

Substituting values:

$$f((P \to \neg Q) \to (Q \vee \neg P)) = \min(1, 1 - 0.9 + 0.2) = \min(1, 0.3) = 0.3$$

Exercise 5: Fuzzy Relations

Consider the following relations, where WGS-5 and WGS-6 stands for Working Group Session 5 and Working Group Session 6, respectively.

Understanding:

	WGS5	WGS6
Joana	0.3	0.7
Andreea	0.6	0.4
Francesca	0.8	0.2

Contains;

	Fuzzy Logic	Probability
WGS-5	0.7	0.3
WGS-6	0.4	0.6

Fill in the following table for the composition of the two relations:

	Fuzzy Logic	Probability
Joana		
Andreea		
Francesca		

Let’s have a look at the given relations!

- **Understanding (Individuals ➡ Sessions):**
  - Joana: $f_{\mathrm{WGS5}} = 0.3$, $f_{\mathrm{WGS6}} = 0.7$
  - Andreea: $f_{\mathrm{WGS5}} = 0.6$, $f_{\mathrm{WGS6}} = 0.4$
  - Francesca: $f_{\mathrm{WGS5}} = 0.8$, $f_{\mathrm{WGS6}} = 0.2$
- **Contains (Sessions ➡ Attributes):**
  - **Fuzzy Logic:** $f_{\mathrm{WGS5}} = 0.7$, $f_{\mathrm{WGS6}} = 0.4$
  - **Probability:** $f_{\mathrm{WGS5}} = 0.3$, $f_{\mathrm{WGS6}} = 0.6$

… DO YOU REMEMBER THE COMPOSITION FORMULA? :)

$$f_{R \circ S}(\langle x, z \rangle) = \max_{y \in Y} \min(f_R(\langle x, y \rangle), f_S(\langle y, z \rangle))$$

- Understanding (Individuals ↔ Sessions):
  - Joana: $f_{WGS5} = 0.3$, $f_{WGS6} = 0.7$
  - Andreea: $f_{WGS5} = 0.6$, $f_{WGS6} = 0.4$
  - Francesca: $f_{WGS5} = 0.8$, $f_{WGS6} = 0.2$
- Contains (Sessions ↔ Attributes):
  - Fuzzy Logic: $f_{WGS5} = 0.7$, $f_{WGS6} = 0.4$
  - Probability: $f_{WGS5} = 0.3$, $f_{WGS6} = 0.6$

Joana:

1. **Fuzzy Logic:** f(Joana, Fuzzy Logic) = max(min(0.3,0.7),min(0.7,0.4)) = max(0.3,0.4) = 0.4
2. **Probability:** f(Joana, Probability) = max(min(0.3,0.3),min(0.7,0.6)) = max(0.3,0.6) = 0.6

Andreea:

1. **Fuzzy Logic:** f(Andreea, Fuzzy Logic) = max(min(0.6,0.7),min(0.4,0.4)) = max(0.6,0.4) = 0.6
2. **Probability:** f(Andreea, Probability) = max(min(0.6,0.3),min(0.4,0.6)) = max(0.3,0.4) = 0.4

Francesca:

1. **Fuzzy Logic:** f(Francesca, Fuzzy Logic) = max(min(0.8,0.7),min(0.2,0.4)) = max(0.7,0.2) = 0.7
2. **Probability:** f(Francesca, Probability) = max(min(0.8,0.3),min(0.2,0.6)) = max(0.3,0.2) = 0.3

Individual	Fuzzy Logic	Probability
Joana	0.4	0.6
Andreea	0.6	0.4
Francesca	0.7	0.3

	toothache		*¬ toothache*	
	catch	*¬ catch*	*catch*	*¬ catch*
cavity	.108	.012	.072	.008
¬ cavity	.016	.064	.144	.576

a) P(cavity, toothache | catch) =
P(cavity & toothache & catch) / P(catch)=
0 .108 / 0.108+0.016+0.72+0.144

b) P(-toothache | cavity) =
P (-toothache & cavity)/ P(cavity) =
0.072+0.008 / 0.108+0.012+0.072+0.008

c) P(-cavity, toothache, catch) = 0.016

	Cold	Trains	Snow	Wind	Classification: Frustrated
D1	Yes	No	No	Yes	No
D2	Yes	Yes	Yes	No	Yes
D3	No	Yes	No	Yes	Yes
D4	No	No	No	No	No
D5	Yes	Yes	No	Yes	No

Apply Naive Bayesian learning with a naïve smoothing of 0.01 on this data to predict whether I will be frustrated on a day where: it is not cold and no train is cancelled although it is windy and it snows.

	Cold	Trains	Snow	Wind	Classification: Frustrated
D1	Yes	No	No	Yes	No
D2	Yes	Yes	Yes	No	Yes
D3	No	Yes	No	Yes	Yes
D4	No	No	No	No	No
D5	Yes	Yes	No	Yes	No

1) $P(F|-C,-T,W,S) = P(F) * P(-C|F) * P(-T|F) * P(W|F) * P(S|F)$

2) $P(-F|-C,-T,W,S) = P(-F) * P(-C|-F) * P(-T|-F) * P(W|-F) * P(S|-F)$

$P(F|-C,-T,W,S) = P(F) * P(-C|F) * P(-T|F) * P(S|F) * P(W|F)$

$P(-F|-C,-T,W,S) = P(-F) * P(-C|-F) * P(-T|-F) * P(S|-F) * P(W|-F)$

These can be estimated from the table:
$P(F) = 2/5$
$P(-F) = 3/5$
$P(-C|F) = 1/2$
$P(-T|F) = 0.01$
$P(S|F) = 1/2$
$P(W|F) = 1/2$
$P(-C|-F) = 1/3$
$P(-T|-F) = 2/3$
$P(S|-F) = 0.01$
$P(W|-F) = 2/3$

$P(F|-C,-T,W,S) = 2/5 * 1/2 * 0.01 * 1/2 * 1/2 = 1/2000 = 0.0005$
$P(-F|-C,-T,W,S) = 3/5 * 1/3 * 2/3 * 0.01 * 2/3 = 4/4500 = 1/1125 = 0.00089$

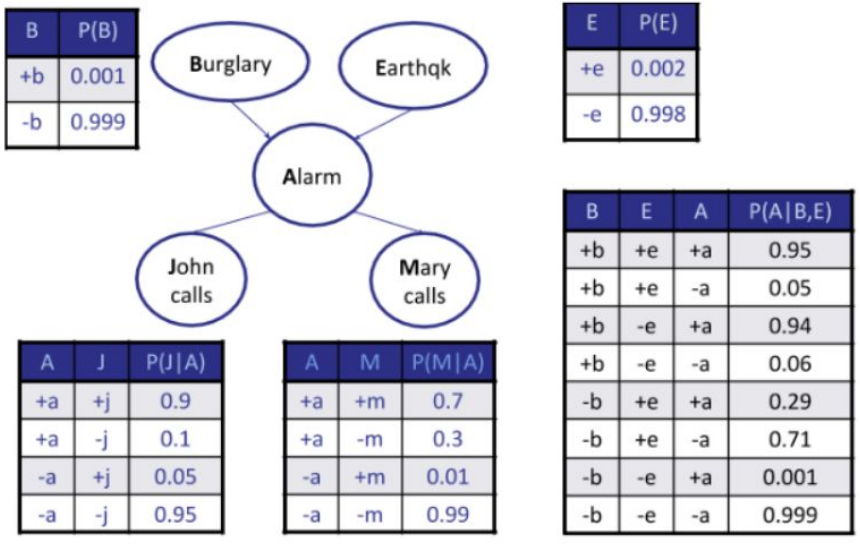
	Cold	Trains	Snow	Wind	Classification: Frustrated
D1	Yes	No	No	Yes	No
D2	Yes	Yes	Yes	No	Yes
D3	No	Yes	No	Yes	Yes
D4	No	No	No	No	No
D5	Yes	Yes	No	Yes	No

P(J,M | A) = ?

J and M are independent =>

P(J|M,A)=P(J|A)* P(M|A)

=0.9*0.7=0.63



B	P(B)
+b	0.001
-b	0.999

E	P(E)
+e	0.002
-e	0.998

B	E	A	P(A|B,E)
+b	+e	+a	0.95
+b	+e	-a	0.05
+b	-e	+a	0.94
+b	-e	-a	0.06
-b	+e	+a	0.29
-b	+e	-a	0.71
-b	-e	+a	0.001
-b	-e	-a	0.999

A	J	P(J|A)
+a	+j	0.9
+a	-j	0.1
-a	+j	0.05
-a	-j	0.95

A	M	P(M|A)
+a	+m	0.7
+a	-m	0.3
-a	+m	0.01
-a	-m	0.99

	actual +	actual -
predicted +	100	20
predicted -	10	30

	actual +	actual -
predicted +	100 True Positives	20 False Positive
predicted -	10 False Negatives	30 True Negatives

Precision = TP/TP+FP = 100 / 120
Recall = TP/TP+FN = 100/110

What will the following sentences become after removing punctuation, stemming and stopwords?

a. Lisa is a student working on Artificial Intelligence.
b. Shuai was feeling lonely.
c. The intelligent systems course had a TA and many students.

a) Lisa student work artificial intelligent
b) Shuai feel lonely
c) Intelligent system course TA many student

Sentence a and c are from the Canvas. Sentence b is from a diary. Considering the following sentence:

a. Lisa is a student working on Artificial Intelligence. - Canvas
b. Shuai was feeling lonely. - Diary
c. The intelligent systems course had a TA and many students. - Canvas

New: There are 200 students enrolled in the Intelligent Systems course.

Using K-NN classification with **hamming distance**, what are the 2 nearest neighbours for this sentence? And thus, what shall be the label if you need to predict the source of this sentence?

Sentence a and c are from the Canvas. Sentence b is from a diary. Considering the following sentence:

a. Lisa is a student working on Artificial Intelligence. - Canvas
b. Shuai was feeling lonely. - Diary
c. The intelligent systems course had a TA and many students. - Canvas

New: There are 200 students enrolled in the Intelligent Systems course.

<Lisa,student,work,artificial,intelligent, Shuai,feel,lone,system,course,TA,many, 200, enrol>

<Lisa,student,work,artificial,intelligent,Shuai,feel,lonely,system,course,TA,many, 200, enrol>

a) <1,1,1,1,1,0,0,0,0,0,0,0,0,0>
b) <0,0,0,0,0,1,1,1,0,0,0,0,0,0>
c) <0,1,0,0,1,0,0,0,1,1,1,0,0,0>

<Lisa,student,work,artificial,intelligent,Shuai,feel,lonely,system,course,TA,many, 200, enrol>

a) <1,1,1,1,1,0,0,0,0,0,0,0,0,0>
b) <0,0,0,0,0,1,1,1,0,0,0,0,0,0>
c) <0,1,0,0,1,0,0,0,1,1,1,0,0,0>

New sentence

d) <0,1,0,0,1,0,0,0,1,1,0,0,1,1>

<Lisa,student,work,artificial,intelligent,Shuai,feel,lonely,system,course,TA,many, 200, enrol>

a) <1,1,1,1,1,0,0,0,0,0,0,0,0,0> - 7/14
b) <0,0,0,0,0,1,1,1,0,0,0,0,0,0> - 9/14
c) <0,1,0,0,1,0,0,0,1,1,1,0,0,0> - 3/14
d) <0,1,0,0,1,0,0,0,1,1,0,0,1,1>

<Lisa,student,work,artificial,intelligent,Shuai,feel,lonely,system,course,TA,many, 200, enrol>

a) <1,1,1,1,1,0,0,0,0,0,0,0,0,0> - 7/14
b) <0,0,0,0,0,1,1,1,0,0,0,0,0,0> - 9/14
c) <0,1,0,0,1,0,0,0,1,1,1,0,0,0> - 3/ 14
d) <0,1,0,0,1,0,0,0,1,1,0,0,1,1>

The closest neighbours are c and a => the new data is coming from Canvas